

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

01

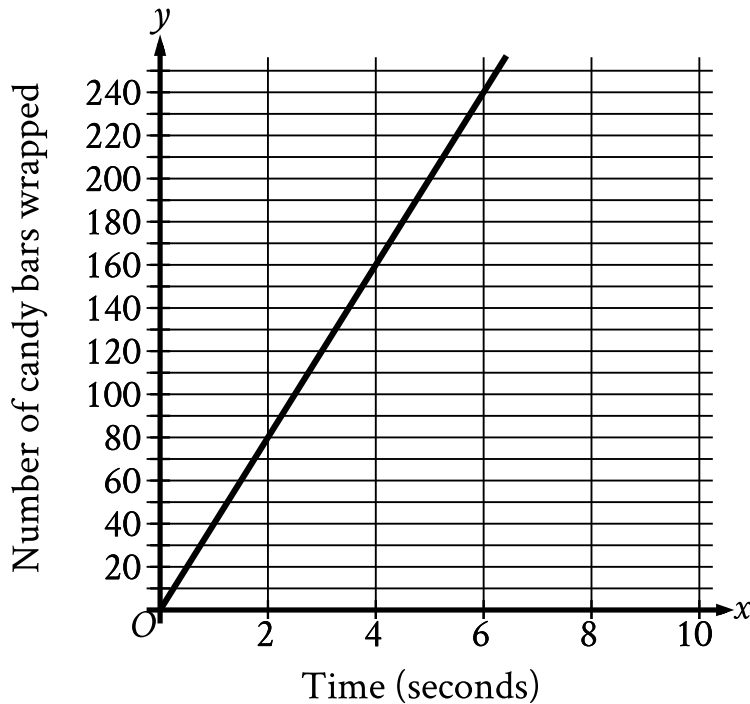
The equation $x + y = 1,440$ represents the number of minutes of daylight (between sunrise and sunset), x , and the number of minutes of non-daylight, y , on a particular day in Oak Park, Illinois. If this day has 670 minutes of daylight, how many minutes of non-daylight does it have?

- A. 670
- B. 770
- C. 1,373
- D. 1,440

If $5x = 20$, what is the value of $15x$?

- A. 7
- B. 12
- C. 23
- D. 60

The graph shown models the number of candy bars a certain machine wraps with a label in x seconds.



- The line slants sharply up from left to right.
- The line passes through the following points:
 - (0 comma 0)
 - (2 comma 80)
 - (4 comma 160)
 - (6 comma 240)

According to the graph, what is the estimated number of candy bars the machine wraps with a label per second?

- A. 2
- B. 40

C. 78

D. 80

A movie theater charges \$11 for each full-price ticket and \$8.25 for each reduced-price ticket. For one movie showing, the theater sold a total of 214 full-price and reduced-price tickets for \$2,145. Which of the following systems of equations could be used to determine the number of full-price tickets, f , and the number of reduced-price tickets, r , sold?

- A. $f + r = 2,145$
 $11f + 8.25r = 214$
- B. $f + r = 214$
 $11f + 8.25r = 2,145$
- C. $f + r = 214$
 $8.25f + 11r = 2,145$
- D. $f + r = 2,145$
 $8.25f + 11r = 214$

A cleaning service that cleans both offices and homes can clean at most 14 places per day. Which inequality represents this situation, where f is the number of offices and h is the number of homes?

A. $f + h \leq 14$

B. $f + h \geq 14$

C. $f - h \leq 14$

D. $f - h \geq 14$

x	y
0	18
1	13
2	8

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A. $y = 18x + 13$
- B. $y = 18x + 18$
- C. $y = -5x + 13$
- D. $y = -5x + 18$

Q8

GRID-IN

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$$10x = 86$$

What value of x is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function g is defined by $gx = 4x - 6$. What is the value of $g-7$?

A. -34

B. -22

C. $-\frac{13}{4}$

D. $-\frac{1}{4}$

Q10

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$$x + y = 125$$

$$x + y + y = 155$$

The solution to the given system of equations is x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.